Kollimarla Sai Sanjay

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience building scalable backend systems with Node.js, TypeScript, MongoDB, Redis and AWS. Experienced in multi-tenant SaaS architecture, asynchronous processing pipelines, database performance optimization and real-time applications, with end-to-end ownership of production software.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, SQL, Python

Backend: Node.js, Express.js, REST APIs, BullMQ, JWT Authentication, WebSockets, Server-Sent Events (SSE)

Frontend: React.js, Angular, HTML, CSS, Bootstrap

Databases: MongoDB, MySQL, Redis

Cloud & Tools: AWS S3, Docker, Git, Postman, Webpack, CI/CD

Core Concepts: Data Structures & Algorithms, System Design, Distributed Systems, Database Design, Caching, Concurrency, Event-Driven Architecture, RBAC, Design Patterns

PROFESSIONAL EXPERIENCE

Software Engineer, NexusIQ Solutions Feb 2024 - Present

- Designed a multi-tenant backend architecture serving multiple organizations with secure tenant isolation and configurable authorization.
- Built RESTful services in Node.js and TypeScript supporting concurrent exam workflows with role-based access control across 40+ permission types.
- Optimized MongoDB queries through indexing, aggregation pipeline improvements and schema redesign, reducing API latency for high-volume operations.
- Built asynchronous pipelines for AI-assisted question generation and evaluation, improving content generation efficiency by over 70%.
- Integrated AWS S3 for scalable media storage; built real-time browser monitoring with WebSockets for low-latency exam supervision.

Full Stack Developer, Freelance May 2023 - Jan 2024

- Delivered end-to-end MEAN/MERN applications for 5+ SME clients, from requirements to deployment.
- Improved API response times by up to 40% through query optimization, indexing and caching.

Junior Software Engineer, Upgrad Insofe Nov 2022 - Apr 2023

- Built Angular components and Node.js APIs for analytics dashboards, reducing page load time by 30% with lazy loading.
- Collaborated with data engineering teams to integrate machine learning models into production APIs.

Machine Learning Intern, Verzeo May 2021 - Jun 2021

- Built an SVM diabetes predictor and a handwritten digit recognizer with 98% accuracy on MNIST using Pandas, Scikit-learn and TensorFlow.

PROJECTS

Production-grade Email Queue Platform
Node.js, TypeScript, BullMQ, Redis, Docker, REST APIs, SSE, Webhooks

- Architected a distributed email processing platform with retries, exponential backoff, dead-letter queues and idempotent processing.
- Built horizontally scalable workers with delayed jobs, priority queues, rate limiting and real-time delivery tracking via SSE and webhooks.

Distributed Rate Limiting Service
Node.js, TypeScript, Redis, Lua, Express

- Implemented Token Bucket and Sliding Window algorithms with Redis atomic operations and Lua scripts to eliminate race conditions; benchmarked latency and scalability.

TinyURL - Distributed URL Shortening Service
Node.js, TypeScript, MongoDB, Redis, Docker, AWS

- Built a scalable URL shortener with click analytics and expiration policies; optimized redirect latency with Redis caching and read-optimized MongoDB indexes.

EDUCATION

Geethanjali College of Engineering and Technology, Hyderabad 2019 - 2023
Bachelor of Technology in Computer Science & Engineering | CGPA: 7.8/10

CERTIFICATIONS

Full Stack Development (React, Node.js) - NxtWave | Java Full Stack - Wipro TalentNext | Machine Learning using Python - Verzeo | Git, SQL, Command Line & Responsive Design - NxtWave